



Studies using concentric ring bifocal and peripheral add multifocal contact lenses to slow myopia progression in school-aged children: a meta-analysis

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Citation information: Li S-M, Kang M-T, Wu S-S, Meng B, Sun Y-Y, Wei S-F, Liu L, Peng X, Chen Z, Zhang F & Wang N. Studies using concentric ring bifocal and peripheral add multifocal contact lenses to slow myopia progression in school-aged children: a meta-analysis. *Ophthalmic Physiol Opt* 2016. doi: 10.1111/opo.12332

Keywords: children, meta-analysis, myopia, soft contact lens

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Received: 25 June 2016; Accepted:
6 September 2016

Abstract

Purpose: To evaluate the effect of soft contact lens with concentric ring bifocal and peripheral add multifocal designs on controlling myopia progression in school-aged children.

Methods: We systematically searched MEDLINE, EMBASE, Cochrane Library and reference lists of included trials. Methodological quality of included trials was assessed using Jadad Scale and Newcastle-Ottawa Quality Assessment Scale items.

Results: We identified five randomised controlled trials (RCTs) and three cohort studies with a total of 587 myopic children. Compared with the control group, concentric ring bifocal soft contact lenses showed less myopia progression with a weighted mean difference (WMD) of 0.31 D (95% CI, 0.05~0.57 D, $p = 0.02$) and less axial elongation with a WMD of -0.12 mm (95% CI, approximately -0.18 to -0.07 mm, $p < 0.0001$) at 12 months. Relative to the control group, peripheral add multifocal soft contact lenses showed less myopia progression with a WMD of 0.22 D (95% CI 0.14~0.31 D, $p < 0.0001$) and less axial elongation of -0.10 mm (95% CI -0.13 ~ -0.07 mm, $p < 0.0001$) at 12 months, respectively. The soft contact lenses with concentric ring bifocal and peripheral add multifocal designs produced additional myopia control rates of 30~38% for slowing myopia progression and 31~51% for lessening axial elongation within 24 months.

Conclusions: Both concentric ring bifocal and peripheral add multifocal soft contact lenses are clinically effective for controlling myopia in school-aged children, with an overall myopia control rates of 30~50% over 2 years. Concentric ring bifocal soft contact lenses seem to have greater effect than peripheral add multifocal soft contact lenses.

Introduction

Over the past few decades, myopia prevalence has increased dramatically worldwide,¹ especially in Asia where it has reached an epidemic level.²⁻⁴ This rapid increase is particularly remarkable in school-aged children who have been

shown a younger age of myopia onset and an increasing prevalence of high myopia.^{1,5,6} Therefore, attempts at myopia control should begin from the stage of school-aged children or even earlier in kindergarten children to prevent future irreversible visual impairment such as myopic retinopathy, which has been ranked the second to third

cause of blindness in adult white populations,⁷ and a leading cause of blindness among Asian adults in China⁸ Taiwan⁹ and Japan.¹⁰

Many attempts have been made to control myopia progression in school-aged children,^{11,12} such as optical corrective devices,^{13–15} pharmacological interventions,^{16,17} and even methods of Traditional Chinese Medicine.^{18,19} However, few strategies have achieved satisfactory results in terms of clinical significance and safety. In previous reports, we found that multifocal spectacles vs single-vision spectacles,¹⁵ atropine vs placebo,¹⁷ orthokeratology vs spectacles,²⁰ could significantly slow myopia progression in school-aged children, and there were greater myopia control effects in Asian children than in white children and in those with higher initial myopia, with similar effect sizes between randomised controlled trials and cohort studies.^{17,20}

Of all these treatment options, soft contact lenses (SCLs) have been evaluated for their effect on slowing myopia progression since the early 1970s.²¹ Some studies reported that SCLs might cause ‘creeping myopia’ compared with spectacles,^{22,23} whereas other studies did not support this.^{13,24,25} In recent years, SCLs with special designs based on new materials and new technologies, such as concentric ring bifocal SCLs^{14,26,27} and peripheral add multifocal SCLs,^{28–32} have attracted great interests in controlling myopia progression in children. The optical design of the concentric ring bifocal SCLs comprises a central distance zone surrounded by a series of treatment/correction zones with near addition. Peripheral add multifocal SCLs, consists of a central zone dedicated to distance vision, surrounded by progressively increased refractive power to reach a relative positive power in the peripheral zones. Myopia control results with these specially designed SCLs were promising but variable. For example, some studies showed a consistent direction of change in both myopia progression and axial elongation,^{14,27,30} whereas others showed an inconsistent direction of changes in myopia progression and axial elongation.^{26,28,29,31,32}

In this article, we aimed to review and meta-analyse evidence from both randomised controlled trials and cohort studies about the effect of SCLs with different designs including concentric ring bifocal and peripheral add multifocal on slowing myopia progression in school-aged children.

Materials and methods

Search strategy

Two reviewers (SML, SSW) independently searched MEDLINE (1966 to May 2016), EMBASE (1950 to May 2016), and Cochrane Library (up to May 2016) using the following keywords: *myopia*, *contact lens*, *child*, *myopia*

progression, *axial length*, *refractive error*, *clinical trial*, and relevant free terms. We used the Boolean operator ‘AND’ to combine all search sets as the final step. Furthermore, we searched the reference lists in the retrieved articles for additional trials, and used Science Citation Index to find studies that had cited the identified trials. Chinese Clinical Trial Registry, WHO International Clinical Trials Registry Platform and ClinicalTrials.gov were also retrieved for ongoing trials. There were no language restrictions in searching for these trials.

Studies selection

Relevant clinical trials were selected according to the following criteria. (1) Study type: given the paucity of available evidence addressing the question of this study and previously reported comparable effect in randomised controlled trials (RCTs) and cohort studies,^{17,20} both RCTs and cohort studies were included, (2) Study participants: 6–18 year old school-aged children with a diagnosis of myopia, (3) Interventions: SCLs with special design in the experimental group, and single vision soft contact lenses or spectacles in the control group since both of them have been reported to have no effects on myopia progression and axial elongation,^{13,24,25} (4) Outcomes: myopia progression (change in spherical equivalent) and axial elongation (change in axial length) at different visits. The exclusion criteria were studies that compared single vision SCLs with spectacles or studies that evaluated gas permeable contact lenses. Two reviewers (SML, SSW) independently assessed the titles and abstracts of all retrieved studies.

Data extraction

Data were independently extracted by two reviewers (SML, SSW) using a customised form with the following information: authors, publication year, country or area, design, length of follow-up, age, baseline refraction, sample size, lost to follow-up, interventions, outcomes, changes in spherical equivalent and axial length, and information on methodology. We contacted the authors of included studies for missing data, if needed, and used GetData Graph Digitizer 2.24 (<http://getdata-graph-digitizer.com>) to read data of different follow-up periods, which were only illustrated in figures.¹⁵

Quality assessment

Quality assessment of all included studies was independently performed by two reviewers (SML, SSW) using the Jadad score for RCTs,³³ which includes assessments of the adequacy of the method for randomisation, blinding procedures, and report of withdrawals and dropouts. Studies that

received an overall score of 3 or higher were classified as high-quality trials. The Newcastle-Ottawa Quality Assessment Scale items³⁴ were used to assess the quality of selected cohort studies. This includes eight items within three domains: selection (representativeness), comparability (because of design or analysis), and outcomes (assessment and follow-up). A study can be awarded a maximum of one star for each numbered item within the selection and outcome categories. A maximum of two stars can be given for comparability. Any discrepancy between the two reviewers about the above issues was resolved by discussion or a third reviewer.

Statistical analysis

Data were entered in Review Manager Version 5.3 (<http://tech.cochrane.org/revman>) for analysis. Differences in myopia progression and axial elongation between two groups were expressed as mean differences (experimental group minus control group, Cochrane Handbook 5.1.0, 9.2.3.1) and 95% confidence intervals (CI). Myopia control rate was defined as the ratio of difference in myopia progression (axial elongation) between two groups and myopia progression (axial elongation) in the control group. Statistical heterogeneity among studies was evaluated using the chi-squared test, and I^2 statistics. A p value less than 0.1 was considered as significant for the test of heterogeneity. A fixed-effect model was used to calculate summary estimates unless there was significant heterogeneity, in which case results were analysed using a random-effect model. Subgroup analyses were pre-planned for the different follow-up periods and interventions. All evaluations were two-way, and a p value of 0.05 was used to declare significance.

Results

A total of 536 publications were retrieved. We excluded 321 duplicate studies, and 207 unrelated studies mainly focusing on different refractive surgeries, cross-sectional studies, and other optical interventions such as contact lenses and orthokeratology. Finally, eight studies published between 1999 and 2016, including five randomised controlled trials^{14,26,27,29,32} and three cohort studies,^{28,30,31} met the inclusion criteria.

Study characteristics

Of the eight studies, three evaluated concentric ring bifocal SCLs.^{14,26,27} Five evaluated peripheral add multifocal SCLs, including two with peripheral gradient SCLs,^{30,31} two with central-distance multifocal SCLs,^{28,29} and one with positive spherical aberration SCLs³² (Table 1). Three were conducted in the USA,^{27,28,32} and one each was conducted in

Table 1. Characteristics of the studies included in the meta-analysis

Study (Author, Year)	Country or area	Study design	Plus power	Follow up (months)	Age (years)	Baseline refraction (D)		Lost/No. of subjects (dropout rate)	
						Experimental	Control	Experimental	Control
Concentric ring bifocal SCLs									
Aller et al. (2016) ²⁷	USA	RCT	+0.25~+3.75 D	12	8~18	-2.57 ± 1.34	-2.81 ± 1.46	1/39 (3%)	0/40 (0)
Anstice et al. (2011) ¹⁴	New Zealand	RCT (Crossover)	+2.00 D	10	11~14	-2.71 ± 1.10	-2.71 ± 1.10	5/40 (13%)	5/40 (13%)
Lam et al. (2014) ²⁶	Hong Kong	RCT	+2.50 D	24	8~13	-2.90 ± 1.05	-2.80 ± 1.03	46/111 (41%)	47/110 (43%)
Peripheral add multifocal SCLs									
Fujikado et al. (2014) ²⁹	Japan	RCT (Crossover)	+0.50 D	24	6~16	-2.56 ± 0.87	-2.64 ± 0.99	0/11 (0)	0/13 (0)
Walline et al. (2013) ²⁸	USA	Cohort study	+2.00 D	24	8~11	-2.24 ± 1.02	-2.35 ± 1.05	13/40 (33%)	13/40 (33%)
Sankaridurg et al. (2011) ³⁰	China	Cohort study	+2.00 D	12	7~14	-2.24 ± 0.79	-1.99 ± 0.62	15/60 (25%)	1/40 (2.5%)
Paune et al. (2015) ³¹	Spain	Cohort study	+6.00 D	24	9~16	-3.76 ± 2.04	-3.11 ± 1.53	11/30 (37%)	20/41 (49%)
Cheng et al. (2016) ³²	USA	RCT	+0.175 µm	12	8~11	-2.52 ± 1.09	-2.44 ± 0.91	12/64 (19%)	6/63 (10%)

SCLs, soft contact lenses.

mainland China,³⁰ Hong Kong,²⁶ New Zealand,¹⁴ Japan²⁹ and Spain.³¹ There were two crossover trials^{14,29} and only the results of the first period were included for analysis.

These eight studies included 587 myopic children, of whom 292 were in the experimental group and 295 in the control group. The length of follow-up period ranged from 10 to 24 months, and the study subjects were aged from 6 to 18 years (Table 1).

Trial quality

The quality of the included RCTs was generally high, with Jadad scores ranging from 4 to 5 (Table 2). All trials reported adequate method of randomisation and numbers of participants lost to follow-up. Only the crossover trial²⁸ did not use allocation concealment due to the characteristic of the design itself. Three trials^{26,27,32} adopted double blinding, two trials^{14,29} adopted single blinding. The quality of the cohort studies was generally high with scores greater than 8 (Table 3).

Follow-up periods and dropout rates

Among the eight studies included (Table 4), four studies had follow-up for 24 months with one of yearly visits,²⁸ 2 of 6-month intervals,^{26,31} and 1 of 3-month intervals²⁹; three studies had follow-up for 12 months with 1 of yearly

visits²⁷ and 2 of 6-months intervals,^{30,32} and the remaining study reported only one follow-up visit at the end of the study of 10 months.¹⁴ The dropout rates ranged from 0 to 41% in the experimental groups, and between 0 to 49% in the control group (Table 1).

Concentric ring bifocal SCLs

The weighted results showed that concentric ring bifocal SCLs yielded less myopia progression than the control group (mean difference, 0.31 D; 95% CI 0.05–0.57 D, $p = 0.02$) at 12 months (Figure 1a). Concentric ring bifocal SCLs also yielded less axial elongation than the control group (mean difference, –0.12 mm; 95% CI approximately –0.18 to –0.07 mm, $p < 0.0001$) at 12 months (Figure 1b).

Peripheral add multifocal SCLs

Compared with the control group, the peripheral add multifocal SCLs had a greater effect on slowing myopia progression of 0.22 D (95% CI 0.14–0.31 D, $p < 0.0001$) at 12 months (Figure 2a). As for slowing axial elongation, the difference was –0.10 mm (95% CI approximately –0.13 to –0.07 mm, $p < 0.0001$) at 12 months (Figure 2b). A stratified analysis by study design showed that the effects were 0.13 D (95% CI –0.02–0.28 D, $p = 0.09$) and –0.11 mm

Table 2. Quality assessment of included randomized controlled trials in the meta-analysis

Study	Randomization	Blinding	Lost to follow up	Allocation concealment	Analysis method	Jadad score
Lam <i>et al.</i> (2014) ²⁶	Adequate	DB	Adequate	Adequate	PP	4
Anstice <i>et al.</i> (2011) ¹⁴	Adequate	SB	Adequate	Adequate	ITT	5
Aller <i>et al.</i> (2016) ²⁷	Adequate	DB	Adequate	Adequate	ITT	5
Cheng <i>et al.</i> (2016) ³²	Adequate	DB	Adequate	Adequate	ITT	5
Fujikado <i>et al.</i> (2014) ²⁹	Adequate	SB	Adequate	Adequate	PP	4

DB, double blinding; SB, single blinding; NO, no blinding or no allocation concealment; PP, per-protocol analysis; ITT, intention-to-treat analysis.

Table 3. Quality assessment of cohort studies included in the meta-analysis using Newcastle-Ottawa Quality Assessment Scale

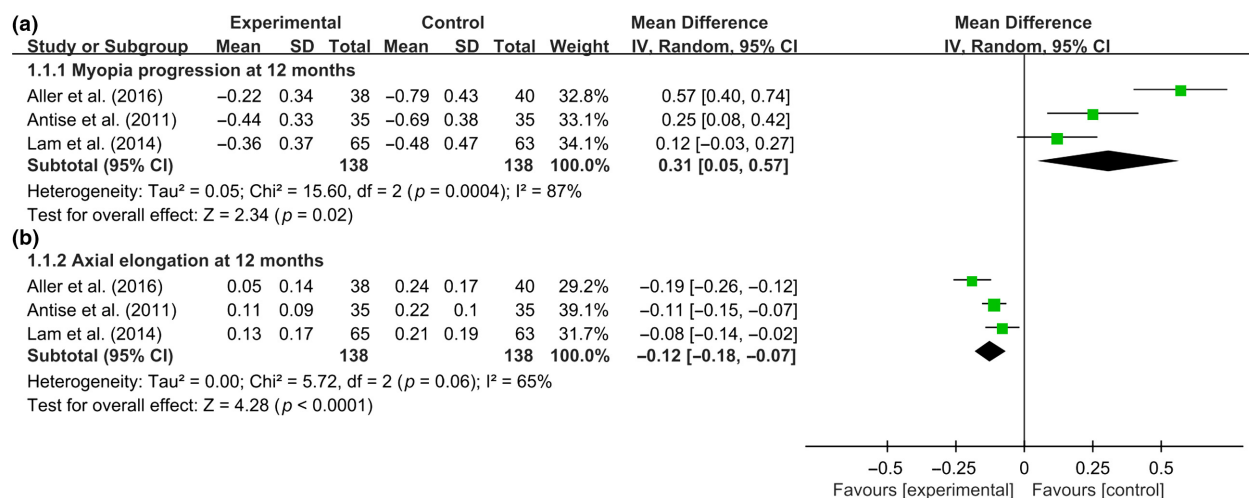
Study	Selection				Outcome				NOS score
	Exposed cohort representative	Non-exposed cohort selection	Exposure ascertainment	Outcome not present at start	Comparability of cohorts	Assessment	Follow-up length	Follow-up adequacy	
Walline <i>et al.</i> (2013) ²⁸	*	*	*	*	**	*	*	*	9
Sankaridurg <i>et al.</i> (2011) ³⁰	*	*	*	*	**	*	*	*	9
Paune <i>et al.</i> (2015) ³¹	*	*	*	*	**	*	*	*	9

Table 4. Myopia progression and axial elongation of included studies during different periods of follow-up

Study (Author, Year)	Follow up (months)	Myopia progression (D)			Axial elongation (mm)		
		Experimental	Control	Control rate	Experimental	Control	Control rate
Concentric ring bifocal SCLs							
Aller <i>et al.</i> (2016) ²⁷	12	−0.22 ± 0.34	−0.79 ± 0.43	72%	0.05 ± 0.14	0.24 ± 0.17	79%
Anstice <i>et al.</i> (2011) ¹⁴	10	−0.44 ± 0.33	−0.69 ± 0.38	36%	0.11 ± 0.09	0.22 ± 0.10	50%
Lam <i>et al.</i> (2014) ²⁶	6	−0.21 ± 0.28 ^a	−0.26 ± 0.33 ^a	19%	0.07 ± 0.11 ^a	0.11 ± 0.12 ^a	36%
	12	−0.36 ± 0.37 ^a	−0.48 ± 0.47 ^a	25%	0.13 ± 0.17 ^a	0.21 ± 0.19 ^a	38%
	18	−0.50 ± 0.43 ^a	−0.71 ± 0.52 ^a	30%	0.20 ± 0.20 ^a	0.29 ± 0.22 ^a	31%
	24	−0.59 ± 0.49	−0.79 ± 0.56	25%	0.25 ± 0.23	0.37 ± 0.24	32%
Peripheral add multifocal SCLs							
Fujikado <i>et al.</i> (2014) ²⁹	12	−0.37 ± 0.33	−0.50 ± 0.18	26%	0.09 ± 0.08	0.17 ± 0.08	47%
Walline <i>et al.</i> (2013) ²⁸	12	−0.33 ± 0.34	−0.60 ± 0.34	45%	0.15 ± 0.17	0.22 ± 0.17	32%
	24	−0.51 ± 0.31	−1.03 ± 0.31	50%	0.29 ± 0.16	0.41 ± 0.16	29%
Sankaridurg <i>et al.</i> (2011) ³⁰	6	−0.28 ± 0.28	−0.57 ± 0.33	51%	0.09 ± 0.11	0.26 ± 0.12	65%
	12	−0.54 ± 0.37	−0.84 ± 0.47	36%	0.24 ± 0.17	0.39 ± 0.19	38%
Paune <i>et al.</i> (2015) ³¹	6	−0.06 ± 0.21	−0.27 ± 0.13	78%	0.13 ± 0.11	0.15 ± 0.09	13%
	12	−0.28 ± 0.38	−0.53 ± 0.25	47%	0.26 ± 0.15	0.28 ± 0.17	7%
	18	−0.39 ± 0.39	−0.80 ± 0.40	51%	0.32 ± 0.19	0.41 ± 0.22	22%
	24	−0.56 ± 0.51	−0.98 ± 0.58	43%	0.38 ± 0.21	0.52 ± 0.22	27%
Cheng <i>et al.</i> (2016) ³²	6	−0.19 ± 0.28 ^a	−0.39 ± 0.33 ^a	51%	0.06 ± 0.10 ^a	0.17 ± 0.10 ^a	65%
	12	−0.55 ± 0.41 ^a	−0.68 ± 0.52 ^a	19%	0.23 ± 0.15 ^a	0.37 ± 0.16 ^a	38%

SCLs, soft contact lenses; Myopia control rate is defined as the ratio of difference in spherical equivalent (axial elongation) between two groups and spherical equivalent (axial elongation) in the control group.

^aMean was derived using GetData Graph Digitizer 2.24 and S.D. was used the greatest S.D. from other studies at the same follow-up period.

**Figure 1.** Concentric ring bifocal soft contact lenses (SCLs) vs the control group on the change in myopia progression (a) and axial elongation (b) at 12 months, respectively.

(95% CI approximately -0.16 to -0.06 mm, *p* < 0.0001) among RCTs, and were 0.27 D (95% CI 0.17–0.38 D, *p* < 0.0001) and -0.09 mm (95% CI approximately -0.14 to -0.04 mm, *p* = 0.0004) among cohort studies (Not shown in figures; *Figure 2*).

Myopia control rate

As for SCLs with designs of concentric ring bifocal, central-distance multifocal, and positive spherical aberration addition, relative to the control group, the summarised myopia

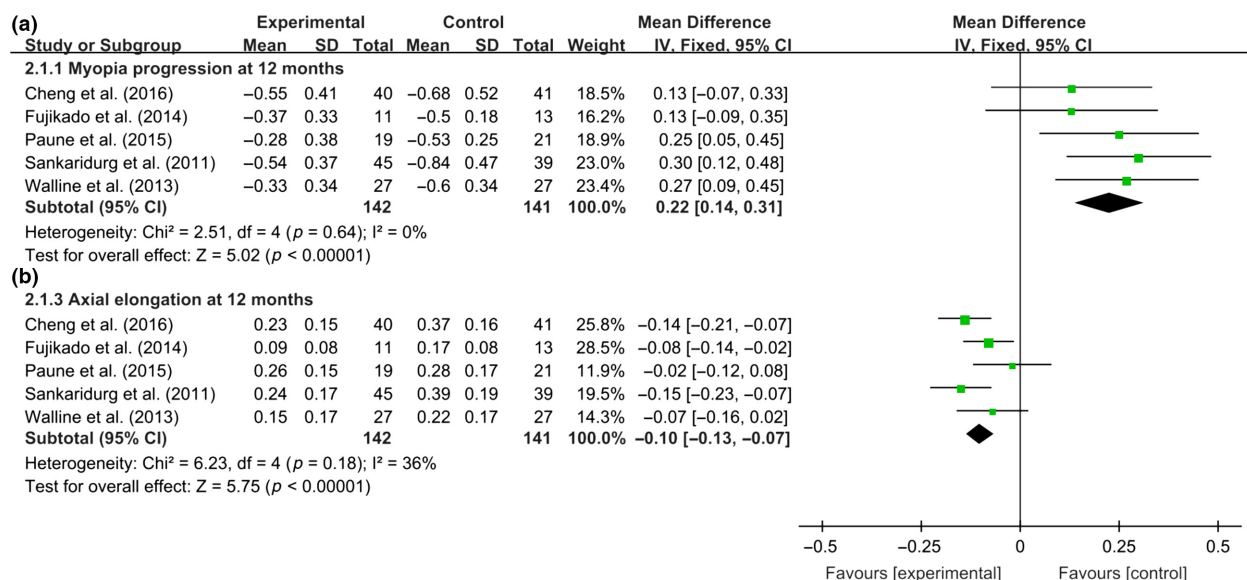


Figure 2. Peripheral add multifocal soft contact lenses (SCLs) vs the control group on the change in myopia progression (a) and axial elongation (b) at 12 months, respectively.

control rate for slowing myopia progression was 35%, 37%, 30% and 38% at 6, 12, 18 and 24 months, respectively, and it was 51%, 47%, 31% and 31% for lessening axial elongation at 6, 12, 18 and 24 months, respectively (Table 4). Compared with the control group, peripheral gradient SCLs had additional myopia control rates for slowing myopia progression of 65% and 42% at 6 and 12 months, and for lessening axial elongation of 39% and 23% at 6 and 12 months, respectively.

Discussion

This meta-analysis summarised currently available evidence from five RCTs and three well-conducted cohort studies that tested the effect of SCLs with different designs on controlling myopia progression in school-aged children aged 6 to 18 years. Compared with the control group, SCLs with concentric ring bifocal and peripheral add multifocal design showed promising results for controlling myopia in school-aged children, with respective effect sizes of 0.31 D and 0.22 D per year for slowing myopia progression, -0.12 and -0.10 mm per year for lessening axial elongation. These myopia control values were slightly smaller than that found with studies using atropine eye drops¹⁷ or orthokeratology.²⁰

When the SCLs had peripheral gradient changes, its advantage over single vision spectacles on slowing myopia progression became prominent with significant differences in myopia progression of 0.24 D at 6 months and 0.28 D at 12 months, respectively, but with no significant difference in axial elongation. These findings support the

evidence from animal studies that modification of peripheral refraction could dominate central refractive development and axial growth.^{35,36} However, other reports suggest that relative peripheral hyperopia does not predict development or progression of myopia in children.^{37,38} Therefore, larger and longer studies on SCLs with change in peripheral refraction to control myopia progression in school-aged children are still needed.

In a previous study,¹⁵ we found that multifocal spectacles slowed myopia progression by a mean refraction of 0.25 D and lessened the axial elongation by a mean of 0.12 mm relative to the single vision spectacles in school-aged children. Interestingly, in the present study on bifocal and multifocal SCLs, we found similar effect sizes on slowing myopia progression (0.22~0.31 D) and axial elongation (0.10~0.12 mm) in school-aged children. These consistent findings suggest that it is the changes in optical characteristic, not the optical devices themselves that affect the refractive development process in school-aged children. However, contact lenses have the advantages of providing greater self esteem for adolescents³⁹, being more convenient⁴⁰ and providing better compliance than spectacles.²⁷

Our previous studies^{15,20} showed that treatment effects of optical interventions were associated with higher baseline myopia and Asian children, and were also associated with lens design features such as aspherical surfaces, foveal defocus for near or distant objects and the magnitude of near addition. We did not perform these analyses in this study because most studies on SCLs had been conducted on white children (5/8) with low myopia on average at baseline (7/8), so that adequate subgroup analyses on these

factors were not possible. As suggested by Walline *et al.*,⁴¹ most of the refractive error treatment effect was limited to the first year of the trial. Although the treatment effect of SCLs may last for 36 months or longer, it still appears to peak during 12–24 months of intervention. Similar results were found in previous studies on multifocal spectacles^{15,42} or RGPs.⁴¹ One important reason may be the optical characteristics of these specially designed SCLs was counteracted by ocular changes. This calls into a question whether the SCLs should be changed every year. Other possible explanations for this transient treatment effect include the corneal flattening induced by contact lenses, the slowing progression rate of myopia with ageing of children, or decreased compliance during treatment. Moreover, few studies have reported any rebound effect from the SCLs after patients had stopped using them. These questions should be further investigated in future trials.

To date, there have been no severe complications reported with SCLs with special designs. According to a recent report, the incidence of microbial keratitis was estimated to be 1–5 per million daily wear soft contact lens users per year,⁴³ which was lower than that of microbial keratitis (13.9% per 10 000 years) in overnight corneal reshaping contact lenses worn by children.⁴⁴ Therefore, the safety of SCLs seems generally high but longer observation in children is required. On the other hand, the decreased retinal image quality and related visual symptoms (especially in low illumination/low contrast conditions) from SCLs with special designs,⁴⁵ should not be ignored and in the future myopia-controlling SCLs should be designed with less compromising of image quality.

New materials and technologies now allow SCLs to deliver drugs or cells to treat ocular and systematic diseases,⁴⁶ and perhaps could be used in the future to deliver low dosage atropine to help myopia control. Furthermore, SCLs allow relatively easy manipulation of light entering the eye, so that controlling the spectral transmittance of the lens is provided. In addition, SCLs are ideal carriers for miniature electronic components, which can be used to monitor important environmental factors such as light levels, reading distance and time spent on different activities.⁴⁷ These applications of SCLs will be important for guiding future myopia interventions and have implications for investigating the mechanisms of myopia pathogenesis.

There are several potential limitations in the present systematic review and meta-analysis. First, the present results were based on data from both randomised controlled trials and cohort studies. Although we have included studies with high quality and these two study designs were found to be comparable in previous studies,¹⁷ the combination of these studies might not eliminate all bias. Second, most studies have been conducted with white children. The treatment effect of SCLs on

controlling myopia in Asian children is expected to be more significant and should be studied in future trials because Asian children typically spend less time outdoors and more time on near work.^{48,49} Third, most of the studies have included children with low myopia and observed them with different follow-up periods.

In conclusion, the present meta-analysis indicates intervention using SCLs with concentric ring bifocal and peripheral add multifocal designs in school-aged children of 6–18 years could significantly slow down myopia progression by about 0.2–0.3 D year⁻¹ and lessen the elongation of axial length by about 0.10 mm year⁻¹, relative to the control group. The treatment effect may last for 24 months or longer. Future randomised controlled trials are expected to evaluate the potential treatment effect of intervention using customised soft contact lens with various durations.

Acknowledgements

This work was supported by the Major International (Regional) Joint Research Project of the National Natural Science Foundation of China (81120108807), the National Natural Science Foundation of China (81300797), Beijing Nova Program (Z121107002512055) and the Major State Basic Research Development Program of China ('973' Program, 2011CB504601). SML and MTK contributed equally to this work.

Disclosure

The authors declare that they have no competing financial interests.

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